

# Mark Opfell

## Skills and Exposure

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| <b>Standards</b>                   | CCSDS, DVB-S2, ITU, FCC, IRIG      |
| <b>RF Tools</b>                    | SDR, VNA, GNU Radio, VSA           |
| <b>General Software Tools</b>      | Python, Git*, Linux, Bash, AWS EC2 |
| <b>Python Libraries</b>            | NumPy, Matplotlib, Scapy           |
| <b>Networking</b>                  | Ethernet, UDP/IP, Wireshark, iPerf |
| <b>FPGA</b>                        | Xilinx, PYNQ, Vivado               |
| <b>Significant Volcano Ascents</b> | Rainier, Baker, Adams              |

## Experience

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| Job Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Senior RF and Telemetry Engineer</b>    |                      |
| Employer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Relativity</b>                          | Long Beach, CA       |
| Period                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>August 2025 – Present</b>               |                      |
| <p>Hands-on debugged a software defined transmitter Xilinx FPGA digital modulator, and frame synchronizer. Corrected, committed, and rebuilt high-level HDL generating code reversing a subtle endianness swap, and non-standard synchronization words. Flashed the output bitstreams onto PCBAs, and validated bit accuracy and frame lock with commercial. Wrote, and tested high-level HDL generating code pseudo-randomization algorithm proving demodulator lock, and basic spectral compliance.</p>                                                                                                        |                                            |                      |
| Job Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>RF Communications System Engineer</b>   |                      |
| Employer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Amazon: Kuiper</b>                      | Redmond, WA          |
| Period                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>July 2024 – August 2025</b>             |                      |
| <p>Developed and ran, over the air Ka-band MIMO phased array system test UDP/IP throughput tests on Xilinx Versal FPGAs for a LEO constellation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                            |                      |
| Job Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Lead Communication Systems Engineer</b> |                      |
| Employer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Albedo</b>                              | Remote & Some Travel |
| Period                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>October 2021 – March 2024</b>           |                      |
| <p>Created, evaluated, and built space-to-ground digital communications payload (gigabit class) for high data rate visible and thermal sensors for VLEO satellites. Developed the telemetry and command data chain from modulated waveform to frames, packets, and connections. Analyzed and tested with: software defined packet radios, GNU radio, technical deep dives into CCSDS standards, and Python code for the processing pipeline.</p> <p>Lead NGSO imaging satellite constellation ITU, and FCC 312 Schedule S regulatory filings. Ran RF analysis efforts with Python scripts, and ITU Spacecap.</p> |                                            |                      |

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|-----------|-----------------------------------|------------------------|
| Job Title | <b>Senior RF Systems Engineer</b> |                        |
| Employer  | <b>BlackSky</b>                   | Tukwillla, WA & Remote |
| Period    | <b>April 2019 – October 2021</b>  |                        |

Created RF architecture diagrams, link budgets, test plans, and ran hands-on troubleshooting. Collaborated with customers and suppliers to design, manufacture, test, launch, and operate X (payload), S (TT&C), GPS, and UHF-band space-based software defined radios linked to ground stations enabled by AWS and the KSAT Lite ground station network.

Collaboratively designed, simulated, sourced, advised layout, and validated: parts, mixed signal PCB, connectors, cabling, and enclosure for a GPS RF system self-compatibility filter. Multiple spacecraft successful in-orbit operation.

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|-----------|-----------------------------------|-------------|
| Job Title | <b>RF Systems Engineer</b>        |             |
| Employer  | <b>Kymeta</b>                     | Redmond, WA |
| Period    | <b>February 2018 – March 2019</b> |             |

Developed and executed over-the-air combined OSI application, transport, network, and physical layer level test cases for a mobile Azure cloud connected MIMO Ku-band terminal with software defined phased array flat panel antennas and a DVB-S2 satellite modem

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|-----------|--------------------------------------|-------------------|
| Job Title | <b>Senior RF Systems Engineer</b>    |                   |
| Employer  | <b>Maxar</b>                         | Mountain View, CA |
| Period    | <b>September 2013 – January 2018</b> |                   |

Wrote specifications, triaged vendors, reviewed test data collateral, and directed the installation, unit level and system level tests of the following passive and active RF units: diplexer, waveguide, directional coupler, band pass filter, low noise amplifier, downconverter, high power load, circulator, coaxial cable, master reference oscillator, and synthesizer.

Developed Python analysis tool from scratch to model complex amplitude and time delay of 10,000+ RF units for ground-based beam-forming.

## Education & Certifications

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|------------|------------------------------------------------------|
| Degree     | <b>Bachelor of Science in Electrical Engineering</b> |
| University | <b>University of California, Davis</b>               |
| Period     | <b>2009 – 2012</b>                                   |

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|---------------|---------------------------|
| Certification | <b>Network Technician</b> |
| Organization  | <b>Cisco</b>              |
| Period        | <b>2024</b>               |

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|---------------|--------------------------------------------|
| Certification | <b>Apprentice Alpine Mountain Guide</b>    |
| Organization  | <b>American Mountain Guide Association</b> |
| Period        | <b>2023</b>                                |

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